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Cretaceous Paleomagnetic Results From Western Tibet and Tectonic Implications

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We present paleomagnetic results obtained from samples collected in the summer of 1989 along a 1000 km traverse between Yecheng (Xinjiang) and Shiquanhe (Tibet) in the western Qinghai-Xizang Plateau (along ~80°E longitude). During this field trip, 480 paleomagnetic cores in Jurassic to Tertiary limestones and sandstones were drilled at 49 sites; 400 were measured. Isothermal remanent magnetization analysis indicates the existence of high coercivity minerals, and thermal, rather than alternating field demagnetization, was found to be more efficient in isolating paleomagnetic directions. The characteristic magnetization has rather low unblocking temperatures (250-560°C) in most limestone samples and higher unblocking temperatures (400°C-680°C) in the sandstone samples. About half of the measured samples show the present geomagnetic field or aberrant directions. The remainder display stable multicomponent magnetizations. The stable characteristic magnetizations indicate (1) a probable Jurassic remagnetization; (2) a Cretaceous (120-80 Ma) magnetization which passes a fold test, yielding a mean pole at 66.2°N, 245.0°E (A_{95}=5.1°, N=14 sites); and (3) a large scatter of Tertiary directions, which may be due to tectonic problems and/or remagnetization in the sampling zone. The Cretaceous paleodirections from Aksaichin and Longmuco (11 sites on the Qiangtang (North Tibet) block) are very similar to those from Shiquanhe (three sites on the Lhasa (South Tibet) block), separated by 300 km. This suggests that the North and South Tibet blocks formed a single unit at least as early as the Cretaceous. Comparing our results with previous paleomagnetic work on the Tibetan plateau, it is found that the paleolatitude of 10.6°±5.1° obtained from this study for a reference at Domar (33.75°N, 80.4°E) is close to that (9.6°±2.8°) obtained by Achache et al. (1984) and Lin and Watts (1988) from the Takena formation (110-100 Ma) in the Lhasa region (30°N, 91°E). In addition, it appears that the southern margin of the Tibetan blocks (i.e., Eurasia) occupied an E-W trending position at about 7°±6°N latitude. Paleodeclination differences between the western and eastern Tibetan blocks suggest internal deformations of the plateau at different scales (from 1 to 1000 km). According to differences in paleomagnetic declinations, the arcuate shape of the blocks would have been partly acquired after the collision.

Introduction

Following the Sino-Swedish expeditions in the early part of this century [e.g., *Henning*, 1915; *Norin*, 1946], interest in the Tibetan plateau was rekindled when it was understood to be a key feature of the India-Asia collision and thereby a prime location for the study of continental deformation [e.g., *Molnar and Tapponnier*, 1975; *Tapponnier et al.*, 1986]. French-Chinese expeditions were organized from 1981 to 1983 [*Tapponnier et al.*, 1981; *Allègre et al.*, 1984], followed by joint British-Chinese and Japanese-Chinese expeditions [*Chang et al.*, 1986; *Otofuji et al.*, 1989]. Following the seminal paper by *Chang and Cheng* [1973], the primary goals of the new studies have been to decipher the number of blocks and sutures that made up the Tibetan plateau as a whole and to determine their tectonic evolution and, in particular, their response to the India-Asia collision.

Paleomagnetic data are among the strongest quantitative constraints that can be sought to reconstruct the block boundaries and their history. Until recently, most data were confined to north-south transects in the eastern part of the plateau, near the longitude of Lhasa [*Pozzi et al.*, 1982; *Westphal et al.*, 1983; *Achache et al.*, 1984; *Besse et al.*, 1984; *Lin and Watts*, 1988]. The one exception is a long largely E-W traverse, along the road from Chengdu to Kashgar [*Otofuji et al.*, 1989]. In the study of *Otofuji et al.* [1989], unfortunately, there were few sites for each stratigraphic interval and few structural constraints, much of the data being gathered from a zone of high strain along the Indus-Zangbo suture. Here, we present the results of sampling along a new traverse (Figures 1 and 2), farther to the west, from Yecheng in Xinjiang province to Shiquanhe in Tibet, on the Indus River, close to the Karakorum fault. Structural data are described elsewhere [P. Matte et al., unpublished manuscript, 1993]. We compare the paleomagnetic results, mainly from Cretaceous rocks, to those that have been obtained in the Lhasa area [*Achache et al.*, 1984] and in other localities in Tibet [*Pozzi et al.*, 1982; *Lin and Watts*, 1988], with the goal of further constraining the finite deformation undergone by the plateau since the onset of the India-Asia collision.

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Geology and Sampling

According to *Chang and Cheng* [1973], the Tibetan plateau consists essentially of four blocks between India to the south and the Tarim block to the north (Figure 1): they include the Lhasa or South Tibet block, the Qiangtang or North Tibet block and the Kunlun and Qaidam blocks in the NE. The blocks are bounded by sutures which are mostly Mesozoic and which

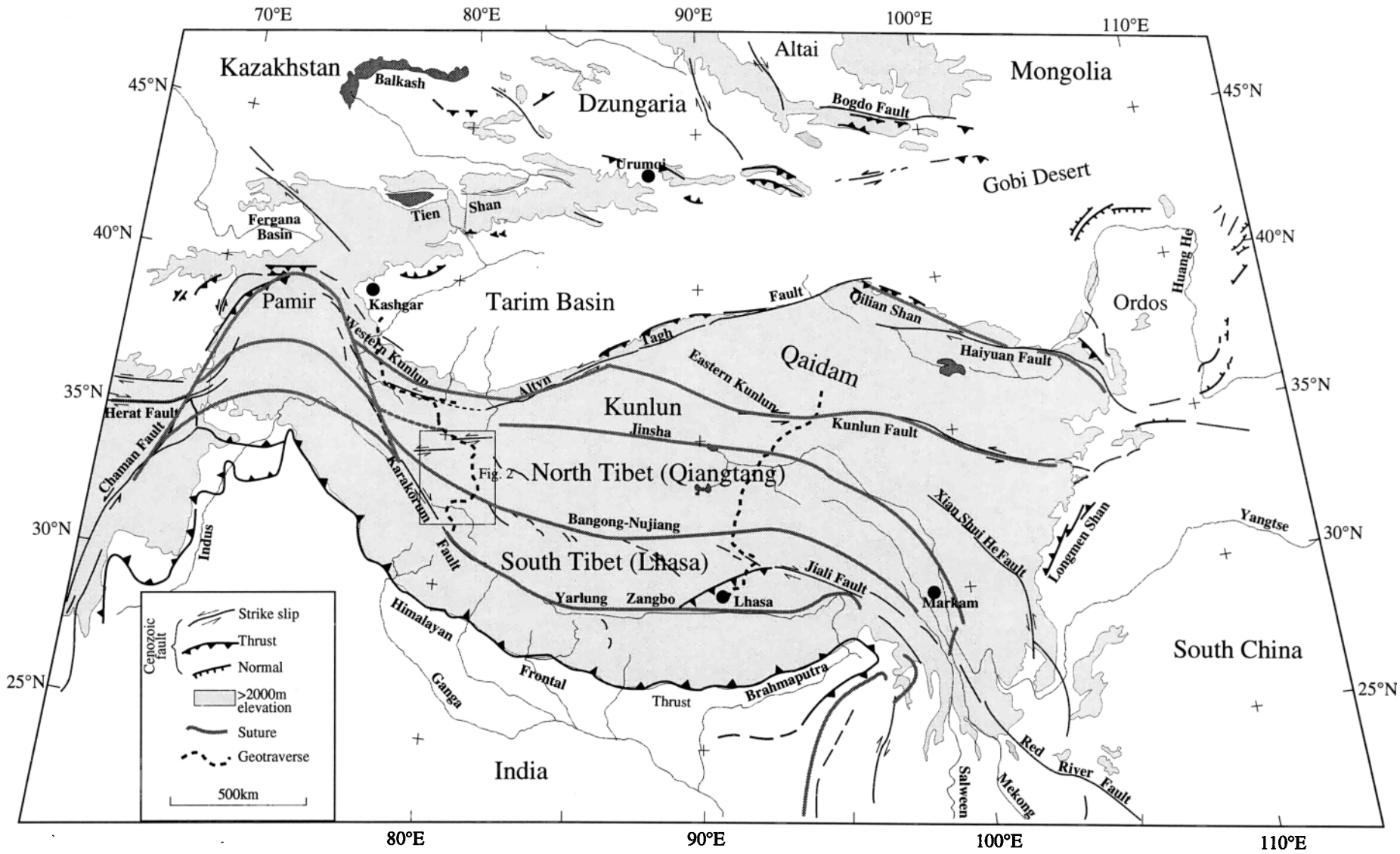


Fig. 1. Simplified tectonic map of Central Asia centered on Tibet, showing sutures and major faults and blocks [after *Avouac*, 1991].

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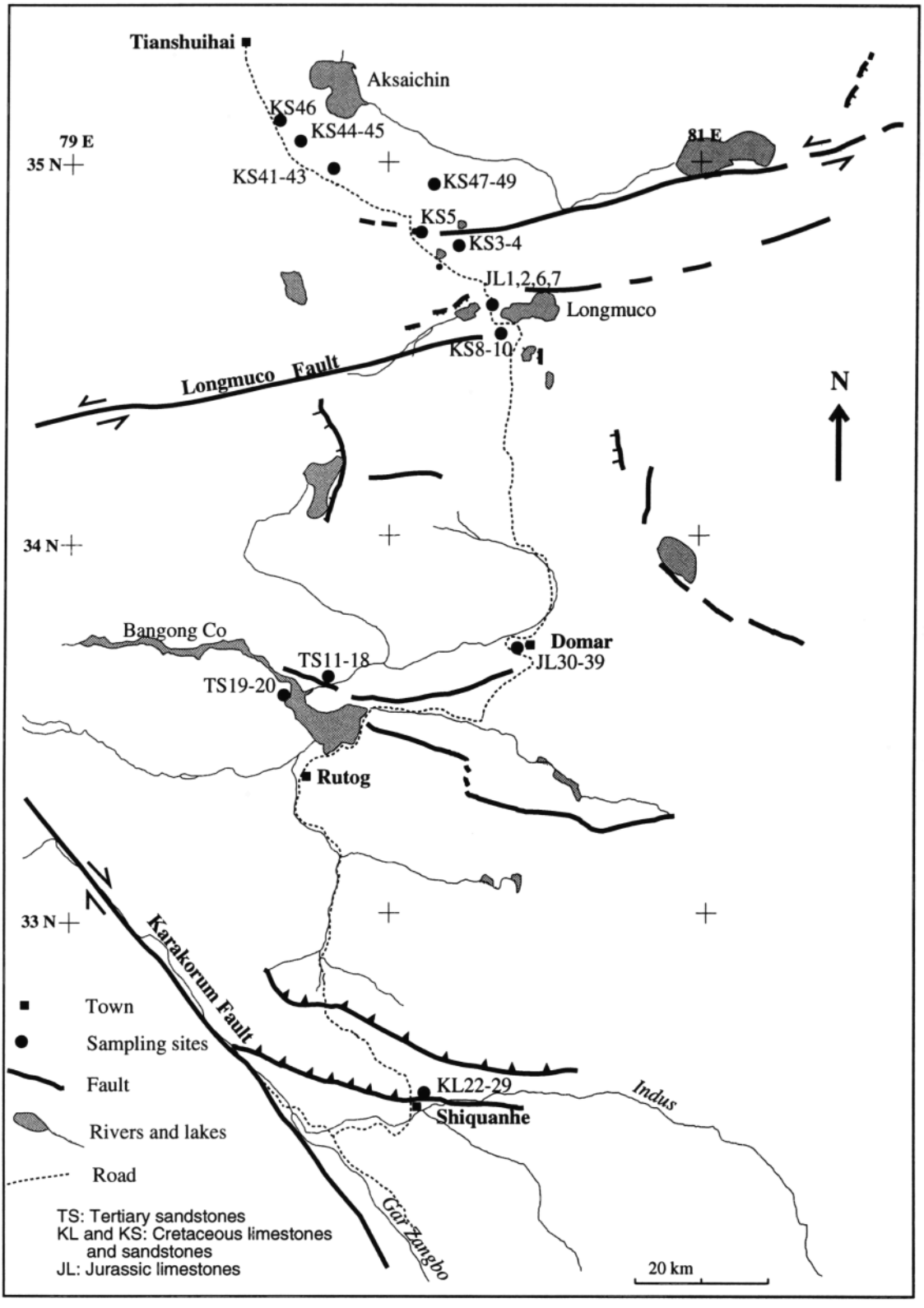


Fig. 2. Simplified map of sampling area, with paleomagnetic sampling sites indicated [after *Bourjot*, 1991].

get older as one progresses northward (although the precise ages of suturing and subsequent reactivation by other collisions are still debated). The Yarlung-Zangbo (Indus) suture separates India from the Lhasa (South Tibet) block and is mostly a Cretaceous to lower Tertiary suture [e.g., *Gansser*, 1964; *Chang and Cheng*, 1973; *Tapponnier et al.*, 1986; *Jaeger et al.*, 1989]; the North and South Tibet (Qiangtang and Lhasa, respectively) blocks are separated by the late Jurassic-early Cretaceous Bangong-Nujiang suture [e.g., *Allègre et al.*, 1984; *Girardeau et al.*, 1984; *Pan and Wang*, 1991; P. Matte et

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al., unpublished manuscript, 1993]; the Qiangtang (North Tibet) and Kunlun-Qaidam blocks are separated by the Jinsha suture of presumed Triassic age [e.g., *Chang and Cheng*, 1973; *Chang et al.*, 1986]. Finally, the Qaidam is separated from the Tarim by the Qilian Shan mountain belt, which represents a reactivated suture of lower Paleozoic age [*Chang and Cheng*, 1973]. These blocks appear to be underlain by Precambrian basement [*Chang et al.*, 1989]. The sedimentary cover on each block ranges from the late Paleozoic to the late Cenozoic. The sampled traverse follows the highway from Tianshuihai (near 35.3°N, 79.6°E) in the north, to Shiquanhe (near 32.5°N, 80.1°E) in the south (Figure 2). The traverse crosses at least three distinct blocks, not all of them with clear-cut boundaries. Detailed descriptions of the geology and tectonics along the traverse are given by P. Matte et al. (unpublished manuscript, 1993) and *Pan and Wang* [1991], and site data are given in Tables 1 and 2.

The traverse begins in the Kunlun block, where the stratigraphic sequence comprises upper Paleozoic limestones, fine-grained terrestrial clastics, and Jurassic coal-bearing clastic sediments. The block is largely intruded by intermediate and acidic magmas, and the sediments have been greatly deformed and metamorphosed [*Pan and Wang*, 1991]. The traverse then crosses into the Qiangtang (North Tibet) block, where the basement is covered by upper Paleozoic formations with Gondwanian facies [*Norin*, 1946; *Wang et al.*, 1981]. Early Upper Permian limestones are unconformably covered with Lower Triassic marls, which in turn are unconformably overlain by Middle Jurassic sandstones and limestones, with a thickness of ~3000 m. The Cretaceous comes in tectonic contact with the Jurassic and is composed of 1500 m of mostly red sandstones with minor reef limestones. The Cenozoic is made of only continental sandstones and conglomerates [*Chang et al.*, 1989]. Ten sites were sampled in the Aksaichin Cretaceous red sandstones (sites KS5 and 41-49; Figure 2). Four sites of Longmuco Jurassic limestones (sites JL 1, 2, 6, 7) and five sites of Longmuco Cretaceous red sandstones (sites KS 3, 4, 8-10) were also sampled in this area.

Ten sites of thin bedded Jurassic limestones (JL 30-39) were sampled near the southern edge of the Qiangtang (North Tibet) block at Domar, and 10 sites of probably Cenozoic red sandstones (TS 11-20) were sampled just north of the Bangong-Nujiang suture, which separates the Qiangtang (North Tibet) and Lhasa (South Tibet) blocks (Figure 2). The basement of the Lhasa block is covered by a Paleozoic and Mesozoic sequence which is well developed near the town of Shiquanhe. Cool-water faunas were found in Carboniferous and Permian limestones [*Wang et al.*, 1981]. Jurassic flysch-type deposits indicate a slope environment with mainly terrestrial sources, and platform-type limestones and plant-bearing shales of Cretaceous age indicate a shallow water continental slope environment [*Pan and Wang*, 1991]. Eight sites were sampled from the orbitolina-bearing Shiquanhe limestones of Cretaceous ages (sites 22-29), about 50 km north of the Karakorum fault where it bounds the Ladakh batholith.

Most age determinations for our samples are based on the paleontological work of the Nanjing Institute of Paleontology. The lithology of the Tertiary sites (TS 11-20) is similar to that of the Cretaceous red beds, though in general coarser, and we did not find fossils that would unambiguously confirm their age; we therefore consider the Tertiary age assignment as tentative. Albo-Aptian (mid-Cretaceous) fossils [e.g., *Eoradiolites gilgitensis*] were found near Tianshuihai [sites KS 44-46; Philippe Matte, personal communication, 1990]. The Cretaceous and Tertiary sites offer the possibility for fold tests, although the precise age of folding is seldom known.

Altogether, 480 cores with 2.5 cm diameter were taken with a gasoline powered drill at 49 sites. The cores were oriented with magnetic and sun compasses. Differences between solar and magnetic azimuths average 1.8°E (σ=4.6°, N=223 samples; D=1.2°, I=51.0° calculated from the 1985 International Geomagnetic Reference Field (IGRF)]. This value

TABLE 1. Paleomagnetic Directions for the High-Temperature Component in Middle Jurassic Sites

Site	s/d	Slat/Slong	n/N	Dg	Ig	Ds	Is	k	α_{95}
01	55/33	34.5/80.4 (LL)	3/5	341.4	44.0	10.4	73.5	131.8	10.8
02	230/20	34.5/80.4 (LL)	3/5	1.4	52.4	353.2	41.6	45.9	18.4
06	100/30	34.5/80.4 (LL)	5/7	3.5	41.4	345.8	77.2	49.5	11.0
07	288/49	34.5/80.4 (LL)	2/5	7.8	71.8	14.6	23.1	-	-
30	320/27	33.8/80.4 (DL)	7/9	40.2	40.0	40.7	14.9	12.4	17.9
31	65/87	33.8/80.4 (DL)	2/6	8.5	43.9	6.2	-39.6	-	-
32	333/45	33.8/80.4 (DL)	5/9	21.3	50.9	37.4	12.4	23.5	16.6
33	55/56	33.8/80.4 (DL)	7/8	41.3	39.5	84.4	30.5	12.5	17.8
34	50/63	33.8/80.4 (DL)	7/7	22.4	49.8	93.5	37.8	59.7	7.9
35	260/38	33.8/80.4 (DL)	6/8	21.9	51.7	10.0	17.1	65.3	8.3
36	165/60	33.8/80.4 (DL)	5/6	10.9	39.0	316.7	37.5	11.6	23.4
37	355/75	33.8/80.4 (DL)	4/6	348.9	40.9	34.1	14.3	113.7	8.7
38	250/40	33.8/80.4 (DL)	3/7	2.3	46.5	355.3	8.3	99.2	12.4
39	225/32	33.8/80.4 (DL)	5/6	358.9	49.0	344.6	22.9	45.7	18.4
Mean			14	10.9	48.5	-	-	30.3	7.3
				-	-	18.8	31.7	4.0	23.0

s/d: strike and dip (in °) of site beds (dips greater than 90° denote inverted beds); Slat/Slong, sampling site coordinates with their location names and lithology (LL, Longmuco limestone and DL, Domar limestone); n/N: respectively, number of specimens used in the statistics and measured; D, I, k, α_{95}: declination, inclination, *Fisher* [1953] precision parameter and radius of 95% confidence cone of site-mean data; subscripts g and s stand for geographic and stratigraphic coordinates respectively; k_s/k_g=0.13;

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TABLE 2. Paleomagnetic Directions for the High-Temperature Component in Cretaceous Sites

Site	s/d	Slat/Slong	n/N	Dg	Ig	Ds	Is	k	α95/A95	Lat	Lon
Results for Individual Sites											
03	108/80	34.5/80.4 (LS)	8/10	198.8	49.2	198.5	-31.0	9.2	19.4	65.7	212.8
04	93/90	34.5/80.4 (LS)	12/12	219.2	72.8	192.8	-13.3	7.1	17.6	59.8	234.4
08	300/30	34.5/80.4 (LS)	9/10	190.5	-43.6	195.5	-14.8	19.5	12.2	59.5	228.9
09	244/85	34.5/80.4 (LS)	12/12	289.8	-54.2	182.5	-29.1	10.7	14.0	70.9	253.6
10[a]	95/99	34.5/80.4 (LS)	9/10	24.6	-83.1	1.9	22.2	3.0	41.6		
21[b]	173/38	32.7/80.2 (SL)	7/7	359.4	52.2	317.7	41.7	221.6	4.1		
22	175/22	32.7/80.2 (SL)	6/7	10.3	30.9	355.5	34.1	26.6	11.9	75.4	277.4
23[b]	215/25	32.7/80.2 (SL)	7/8	0.8	46.1	346.1	29.2	51.7	8.6		
24[b]	95/26	32.7/80.2 (SL)	4/6	358.3	47.1	349.5	72.7	110.5	9.3		
25[b]	180/7	32.7/80.2 (SL)	2/6	357.8	47.1	350.4	46.4	-	-		
26[b]	257/60	32.7/80.2 (SL)	5/5	8.7	50.1	0.9	-7.7	251.0	4.8		
27[b]	250/45	32.7/80.2 (SL)	5/7	345.8	61.8	301.7	40.4	67.0	9.4		
28	205/25	32.7/80.2 (SL)	8/9	21.3	16.7	14.5	13.5	18.1	13.4	60.8	229.5
29	250/25	32.7/80.2 (SL)	8/12	28.1	40.5	17.6	22.0	38.3	9.1	63.3	219.0
40[b]	358/20	35.0/79.7 (AS)	3/8	358.5	45.8	17.7	42.2	81.9	16.1		
41	160/15	35.0/79.7 (AS)	9/11	0.0	18.8	354.4	23.2	19.3	12.0	66.5	273.6
42	150/12	35.0/79.7 (AS)	5/10	0.9	12.9	357.9	18.8	9.7	26.5	64.6	264.5
43[b]	105/25	35.0/79.7 (AS)	5/7	352.6	48.9	336.3	69.8	25.3	15.5		
44	198/17	35.0/79.7 (AS)	6/8	2.3	27.8	356.8	20.4	34.3	11.8	65.4	267.3
45	213/15	35.0/79.7 (AS)	4/7	18.2	22.8	16.1	11.5	43.0	16.4	52.6	232.5
46	200/12	35.0/79.7 (AS)	9/11	6.5	25.6	0.7	21.7	31.0	9.5	66.2	258.0
47[a]	65/12	35.0/79.7 (AS)	6/9	148.6	-56.9	145.3	-68.7	8.6	30.7		
48	125/15	35.0/79.7 (AS)	3/8	0.7	17.3	356.9	29.4	33.1	21.8	70.5	268.7
49	310/15	35.0/79.7 (AS)	5/8	0.1	34.7	5.1	22.8	17.7	18.7	66.4	247.1
Mean of All Site Directions (excluding a and b sites)											
			14	13.3	20.7	-	-	4.5	21.1		
				-	-	6.2	22.1	53.2	5.5		
Mean of Individual Poles and Recalculated Directions for a Reference Site at Domar (33.75°N, 80.4°E)											
			14	17.3	8.7	-	-	5.5	18.8	56.5	227.9
				-	-	6.3	20.6	61.9	5.1	66.2	245.0
Averages for Separate Sampling Areas											
Aksaichin			7	4.1	23.0	-	-	73.0	7.1		
				-	-	1.3	21.8	83.6	6.6	66.3	256.5
Longmuco			4	218.4	6.9	-	-	1.3			
				-	-	192.4	-22.2	52.4	12.8	64.4	231.3
Shiquanhe			3	19.6	29.6	-	-	32.8	21.9		
				-	-	9.8	23.5	29.9	23.0	67.7	234.2

As in Table 1 (LS, SL and AS are Longmuco sandstone, Shiquanhe limestone and Aksaichin sandstone, respectively); A95: radius of 95% confidence cone around the mean pole; Lat, Lon: latitude and longitude of (virtual) paleomagnetic pole calculated from the site direction Ds, Is.

[a] Sites showing aberrant directions which are excluded from the mean direction.

[b] Sites showing a direction close to the present earth field in geographic coordinates, and rejected from the mean direction.

was used to correct the declination of all magnetically determined orientations and bedding strikes.

PALEOMAGNETIC MEASUREMENTS

Magnetic measurements were performed with the CTF three-axis cryogenic magnetometer in the shielded laboratory at the Institut de Physique du Globe in Paris. Alternating field (AF) demagnetization, using a Schonstedt demagnetizer, proved for most part to be ineffective; therefore, most samples were thermally demagnetized using field-free (less than 10 nT) laboratory-built furnaces. Ten to 15 heating steps were made, and the bulk susceptibility of each specimen was monitored after each temperature step in order to check for magnetic mineral changes. Isothermal remanent magnetization (IRM) experiments were carried out on several specimens of different age and rock type.

Demagnetization results were plotted as orthogonal vector diagrams and stereographic projections. Paleomagnetic directions and planes were determined by principal component

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analysis [*Kirshvink*, 1980] and site means from combined data were determined by McFadden's method [*McFadden and McElhinny*, 1988].

Middle Jurassic Sites

As noted above, AF demagnetization was ineffective for most samples. Figures 3*a* and 3*b* show an example from the Domar Jurassic limestones, in which the AF technique removes a significant overprint in the direction of the present earth field (PEF), leaving a characteristic reversed magnetization with coercivities above the 100 mT peak field. The magnetic minerals are better demagnetized with thermal treatment. Figures 3*c*-3*k* show three examples of Jurassic limestones from Domar and Longmuco, which display one or two magnetic components. A lower unblocking temperature component (LTC) is clearly present in Figures 3*c*-3*e*, with 80% of the magnetization removed by 250°C. Seventy percent of the magnetization is removed by 320°C in Figures 3*f*-3*h*, whereas there is no such behaviour in Figures 3*i*-3*k*. The LTC could be carried by goethite or sulfides, or have a viscous origin. The Fisherian mean direction for 53 samples is D=3.7°, I=50.7° (α_{95}=3.3°) in geographic coordinates and is very close to the present field (D=1.2°, I=51.0°) and the recent axial-dipole field direction (D=0°, I=53°). A fold test for this component is clearly negative, with a decrease of the precision parameter k by a factor of 11 upon tectonic correction.

The higher-temperature components (HTC) unblock between 250°C and 650°C. Some components display almost univectorial behavior up to the highest thermal step such as the sample shown in Figure 3*k* where the maximum unblocking temperature (Tub) is above 645°C. IRM acquisition curve (Figure 3*j*) confirms the presence of hematite in these samples. In contrast, the Tub and coercivity of the sample shown in Figures 3*f*-3*h* is characteristic of magnetite. The sample in Figures 3*c*-3*e*, with a Tub of 400°C and low coercivity might correspond to (titano) magnetite or a sulfide. The mean of the HTC components from the 14 Jurassic sites is D=10.9°, I=48.5° (α_{95}=7.3°) in geographic coordinates, again close to the PEF (Table 1). The fold test is negative, with the ratio of the precision parameters before and after bedding correction in excess of 7.5 (Figure 4). It therefore appears that the Jurassic rocks from the Qiangtang (North Tibet) block have been almost entirely remagnetized in fairly recent times and that these data cannot be used for paleogeographic reconstructions.

Cretaceous Sites

Twenty-four Cretaceous sites, separated by up to 300 km, were collected near Shiquanhe (limestone, Figures 5*a*-5*f*), Longmuco (red sandstones, Figures 5*g*-5*j*) and Aksaichin (red sandstones). Again, thermal demagnetization led to the identification of a LTC and a distinct HTC. The LTC was usually removed by 200°C and represented on average 40% of the initial natural remanent magnetization (NRM) in the Aksaichin sandstones and Shiquanhe limestones (Figures 5*a* and 5*d*) and 20% in the Longmuco sandstones (Figure 5*g*). The statistics for these LTCs from 123 specimens confirm a PEF overprint (D=355.5°, I=51.4°; α_{95}=2.4°) with a strongly negative fold test (k being 8.8 times larger in geographic than in stratigraphic coordinates).

The Tub for the HTC varies with rock type, as expected, being generally lower in the limestones than in the sandstones. Figures 5*g*-5*j* show results from a typical red sandstone from Longmuco, with a characteristic component isolated between 450 and 670°C. The maximum Tub and high coercivity (Figure 5*h*) are characteristic of hematite. A decrease in bulk susceptibility after 300°C, and particularly after 600°C does not appear to affect the magnetic direction (Figures 5*i* and 5*j*). In contrast, the limestone sample from Shiquanhe shown in Figures 5*a*-5*c* appear to be dominated by a (titano) magnetite, with maximum Tub near 520°C and slight spurious remagnetization beyond. The IRM experiment (Figure 5*b*) reveals both low and high coercivities, the latter possibly linked with the strong LTC (goethite?). Sometimes, mineral changes occur beyond 500-550°C in these limestones (Figure 5*e*), leading to spurious remagnetization at the high-temperature end of the demagnetization diagram (Figure 5*f*).

In some cases, the limestones showed essentially univectorial PEF directions (Figure 5*k*), whereas some sandstones showed erratic behavior upon demagnetization (Figure 5*l*). Unfortunately, 50% of the samples had either of these two unusable behaviors. In 14 out of 24 sites, a HTC could be isolated: four with reversed polarity near Longmuco, three near Shiquanhe and seven near Aksaichin with normal polarity. All site mean directions are displayed in Figure 6. The random type sites are shown as squares and have very large uncertainties. The PEF remagnetization type are shown as triangles; they tend to be well determined and, of course, cluster near the PEF in geographic coordinates, with a negative fold test. The remaining characteristic directions, shown as circles, form two antipodal clusters and indicate a positive fold test. Because the corresponding sites are separated by about 300 km in latitude (Figure 2), we prefer to display the results as poles (Figure 7).

We have calculated, both in terms of directions and poles, the site mean averages for the three separate sampling areas (Table 2 and Figure 8). The angular distances between pairs of mean directions are 2.7° between Longmuco and Shiquanhe, 8.0° between Aksaichin and Shiquanhe, and 10.3° between Aksaichin and Longmuco. The maximum distance is 10.3°±14°, smaller than the minimum critical value (11.6°) for the *McFadden and McElhinny* [1990] reversal test (at Rc level) and not significantly different from zero. We have therefore calculated a single mean for all poles from the three areas, at the same time evaluating a fold test (Table 2 and Figure 7). This test is positive at the 99% confidence level (N=14, k_S/k_G=11.3>2.6 [*McElhinny*, 1964]). The overall Cretaceous mean pole is at 66.2°N, 245.0°E (A_{95}=5.1°, N=14). The corresponding mean direction reduced to Domar (33.8°N, 80.4°E) is D=6.3°±5.2°, I=20.6°±5.1° (Table 2).

The dual polarities and positive fold test and diverse rock types (limestones and sandstones) and magnetic carriers (magnetite and hematite) lead us to propose that we have indeed succeeded in isolating a primary (mid-) Cretaceous direction. The age of this magnetization is partly constrained by the observation of Albo-Aptian fossils in the northernmost Aksaichin area and near Shiquanhe. However, the dual polarities imply that the age is not confined to the Cretaceous long normal superchron, as appeared to be the case in the roughly coeval Takena red sandstones in the Lhasa area [*Achache et al.*, 1984]. If the Tibetan blocks were relatively stable with respect to Eurasia in Cretaceous times, the ages of magnetization in the areas we sampled could be slightly different yet the magnetic directions could be similar, since Eurasia was essentially stable in terms of both latitude and orientation at the time of the "hairpin turn" in the Eurasian apparent pole wander path (APWP), from 130 to 70 Ma [*Besse and Courtillot*, 1991].

The consistency of the directions from three separate areas, spanning 300 km and two separate blocks (North and South Tibet) implies that the blocks have formed a single unit since the time of magnetization. This is in agreement with the Upper Jurassic to Lower Cretaceous age proposed for the suture farther east [e.g., *Girardeau et al.*, 1984; *Pan and Wang*, 1991]. It also implies that the blocks did not suffer significant

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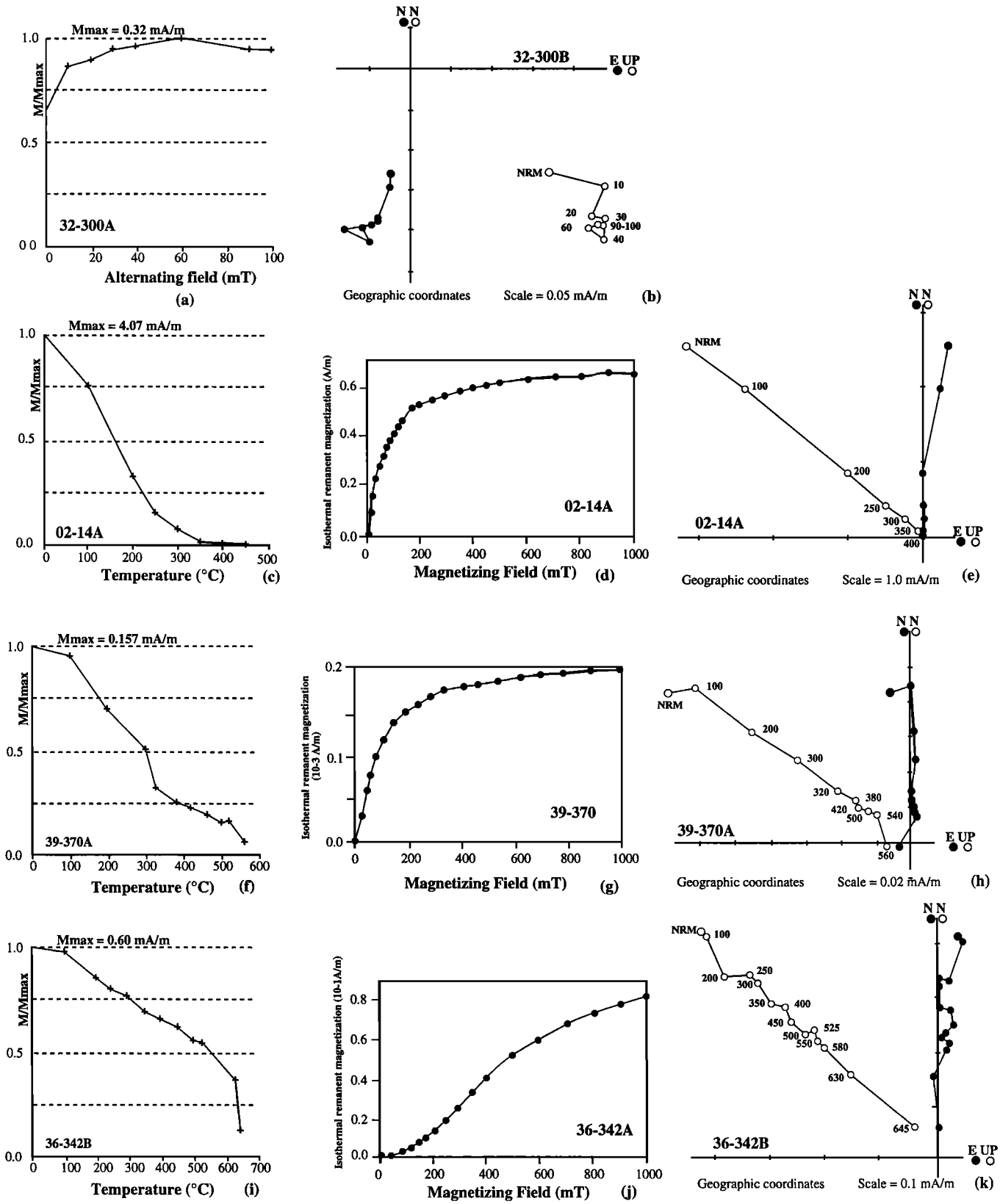


Fig. 3. Results of magnetic study for Jurassic samples. (*a*) and (*b*) Normalized natural remanent magnetization (NRM) intensity curve, and orthogonal projection of AF demagnetization for a Domar limestone showing high-coercivity behavior (demagnetization steps in millitesla). (*c*), (*d*), and (*e*) Thermal demagnetization and isothermal remanent magnetization (IRM) acquisition for a Longmuco limestone showing a large component with lower unblocking temperature and presence of both low- and high-coercivity magnetic minerals. (*f*), (*g*), and (*h*) Same for some Domar limestone showing a magnetic behavior similar to that of the Longmuco limestones. (*i*), (*j*), and (*k*) Same for some Domar limestones showing the presence of high-unblocking temperature and high-coercivity minerals.

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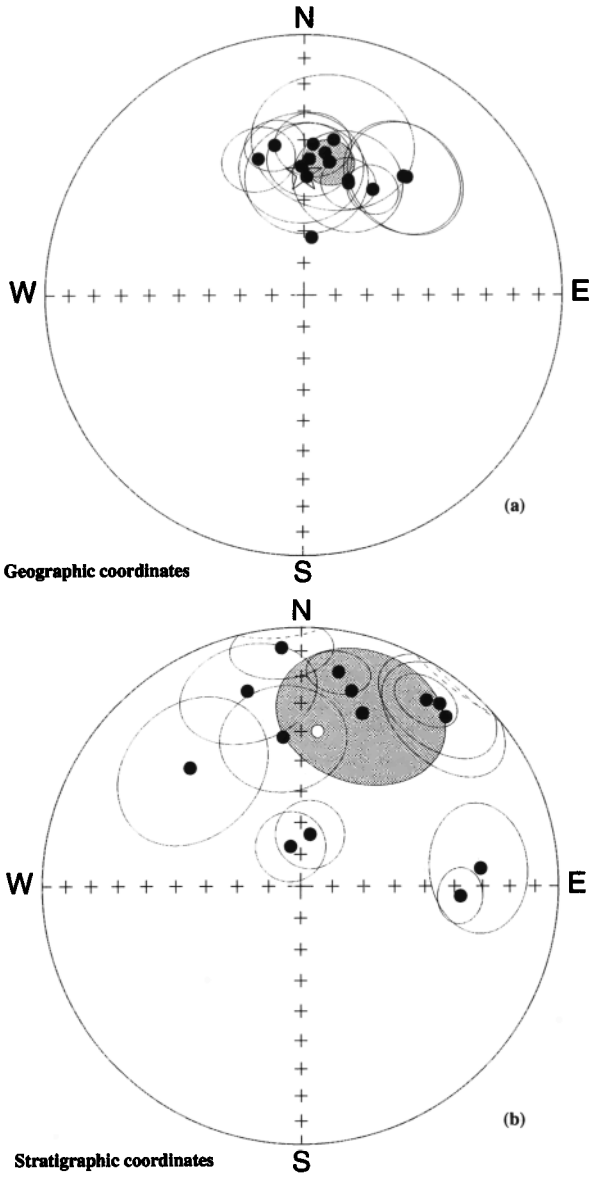


Fig. 4. Equal-area projection of site-mean directions of high temperature component from Jurassic sites (*a*) before and (*b*) after bedding corrections. The 95% confidence oval of the overall mean direction is shaded. Note that the directions are close to the present earth field (PEF, shown by star) in geographic coordinates. Solid (open) symbols are for directions in lower (upper) hemisphere. Symbols without α_{95} confidence oval correspond to sites with less than three samples.

internal deformation at the scale of our sampling. At the reference site in Domar (33.75°N, 80.4°E), the (mid-) Cretaceous paleolatitude is rather low, at 11°±5°, with a slight anticlockwise rotation of 6.6°±6.0° compared to the Eurasian reference [*Besse and Courtillot*, 1991].

Tertiary

Thermal demagnetization isolated a stable high temperature characteristic component in most specimens. However, these components show a very large scatter at the within-site as well as between-site scales, which may be due to tectonic or remagnetization problems. These results are discussed in detail elsewhere [*Chen*, 1992].

PREVIOUS PALEOMAGNETIC STUDIES IN TIBET

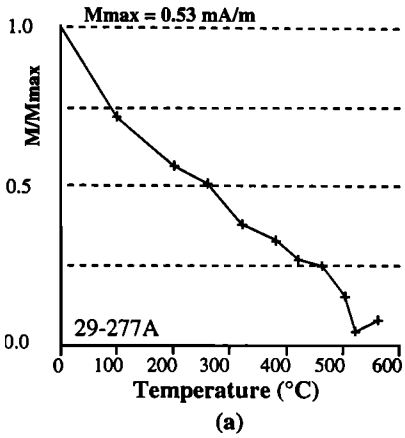
Paleomagnetic studies in the Tibetan plateau began by the end of the 1970s [*Zhu et al.*, 1977; *Zhu et al.*, 1981], with a focus on Cretaceous red sandstones; because AF demagnetization was often preferred, significant characteristic directions could not be extracted and the published directions may not be reliable. Figure 12 and Tables 3 and 4 summarize previous paleomagnetic studies from the various blocks of the Tibetan plateau.

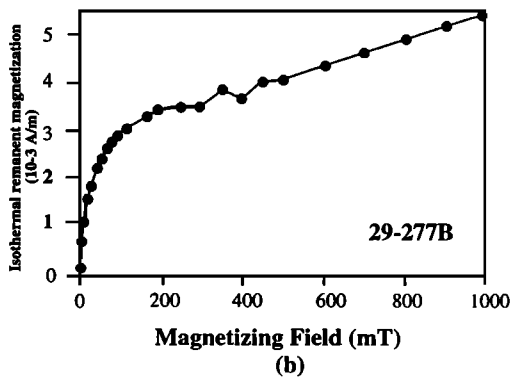
Pozzi et al. [1982] and *Westphal et al.* [1983] reported results from the first French-Chinese (1980) expedition near Lhasa: they sampled the strongly folded Takena Cretaceous red sandstones and the andesites of the unconformably overlying Paleocene Lingzizong formation. *Pozzi et al.* [1982] pointed out that six out of eight poles for the Takena formation aligned with the APWP of India, suggesting the possibility of successive remagnetization. *Westphal et al.* [1983] isolated characteristic directions giving a paleolatitude of 21°±6°N for the Lhasa area. *Achache et al.* [1984] extended sampling during the second French-Chinese (1981) expedition and concluded that the primary component had not been fully separated from steeper secondary magnetizations in the previous studies. *Achache et al.* [1984] proposed a paleolatitude of 13°±3°N for the Lhasa area. When analyzing the paleomagnetic data from the third French-Chinese (1982) expedition, *Besse* [1986] reviewed the data and concluded that the Lhasa area was indeed located between 10° and 15°N in the middle Cretaceous.

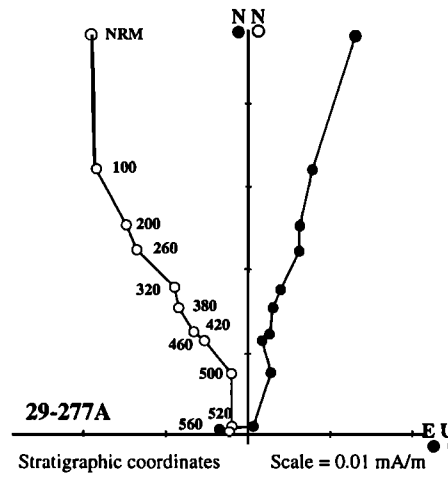
In 1985, a British-Chinese expedition was undertaken along the highway from Lhasa to Golmud, extending into the Qiangtang (North Tibet) and Kunlun-Qaidam blocks, whereas the three French-Chinese expeditions had been confined to the Indian, Lhasa (South Tibet) and southernmost part of Qiangtang (North Tibet) blocks (Figure 1). *Lin and Watts* [1988] presented Cretaceous paleomagnetic data for the Lhasa (South Tibet) block, Triassic, Jurassic, and Tertiary data for the Qiangtang (North Tibet) block and Carboniferous and Triassic data from the Kunlun block. Based on ages and sampling areas, Lin and Watts proposed 10 separate sets of results. Only one group, sampled in the Takena red beds in the Linzhou area showed a positive fold test, similar to the results of *Achache et al.* [1984]. The other data sets provided either no fold test or even negative fold tests. For instance, data from two nearby areas near Erdaogou on the Qiangtang (North Tibet) block, in the same Paleocene-Eocene Fenghoushan Formation, yielded inconsistent results, with respective paleolatitudes of 7°±4° N and 18°±3°N; partial later remagnetization is likely since some of their "primary" directions appear to be very close to the PEF in geographic

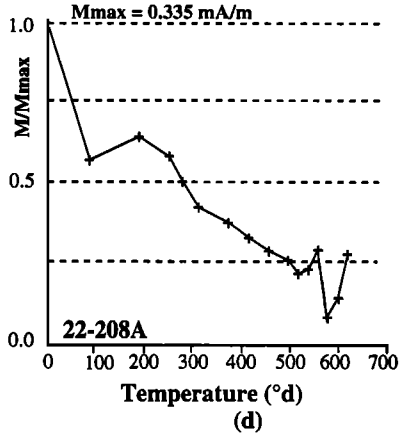
Fig. 5. Results of magnetic study for Cretaceous samples. (*a*), (*b*), and (*c*) Normalized NRM intensity curve, acquisition of IRM and representative orthogonal vector projection for Shiquanhe limestones showing that (titano) magnetites are the characteristic carrier, though both high- and low-coercivity minerals are present. (*d*), (*e*), and (*f*) Normalized NRM, bulk susceptibility and representative orthogonal vector projection for Shiquanhe limestones showing mineral changes beyond 500-550°C leading to spurious remagnetization. (*g*), (*h*), (*i*), and (*j*) Normalized NRM, IRM, bulk susceptibility, and representative orthogonal vector projection for Longmuco red sandstone showing that hematite is a likely carrier of characteristic magnetization (bulk susceptibility changes have a negligible effect on magnetic direction). (*k*) and (*l*) Orthogonal projections showing univectorial PEF and erratic magnetic behavior for some of the Cretaceous samples.

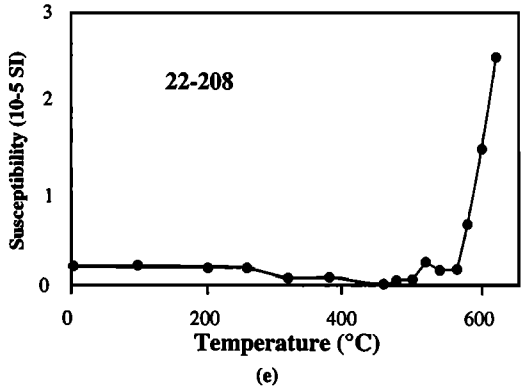
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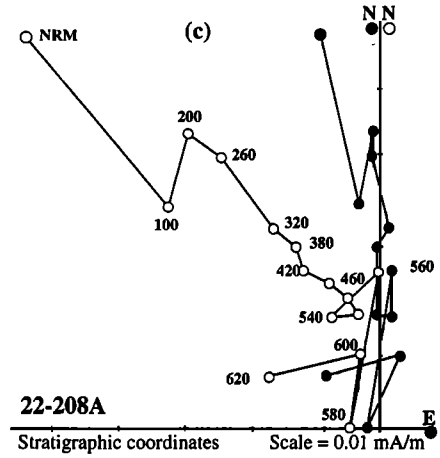


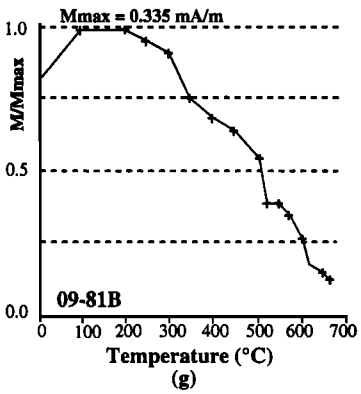


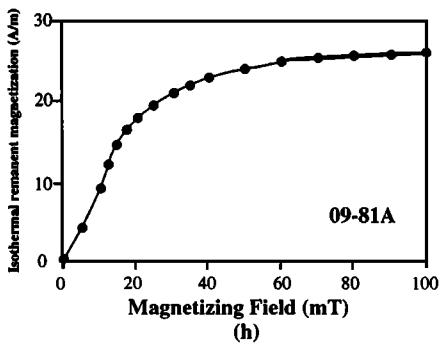


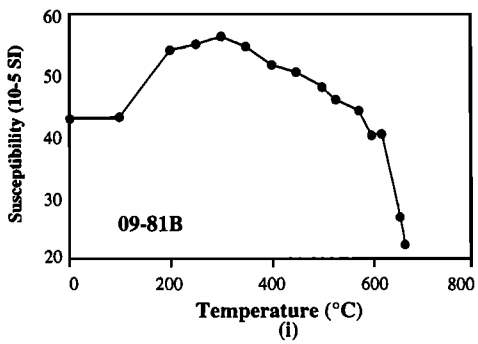


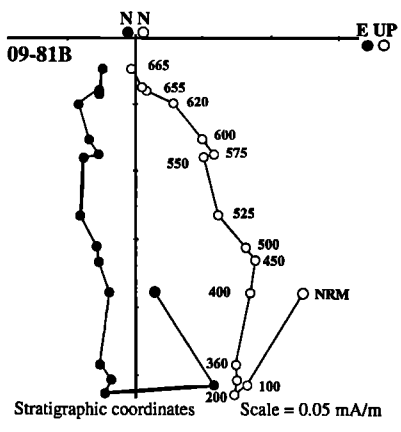


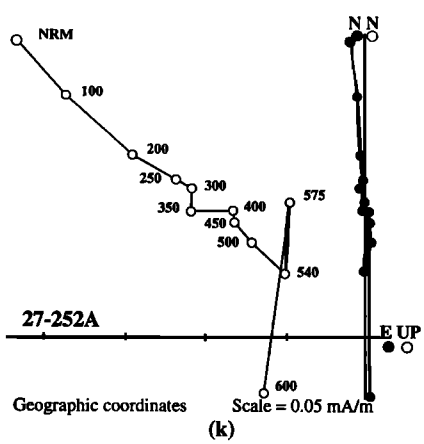


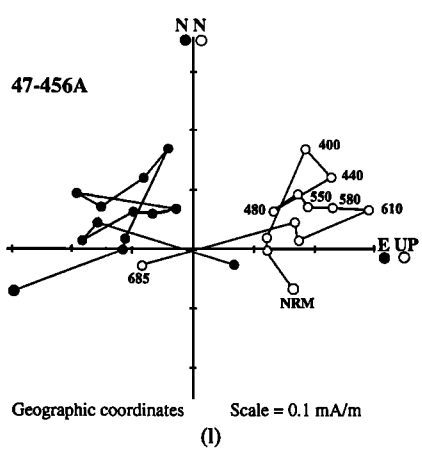












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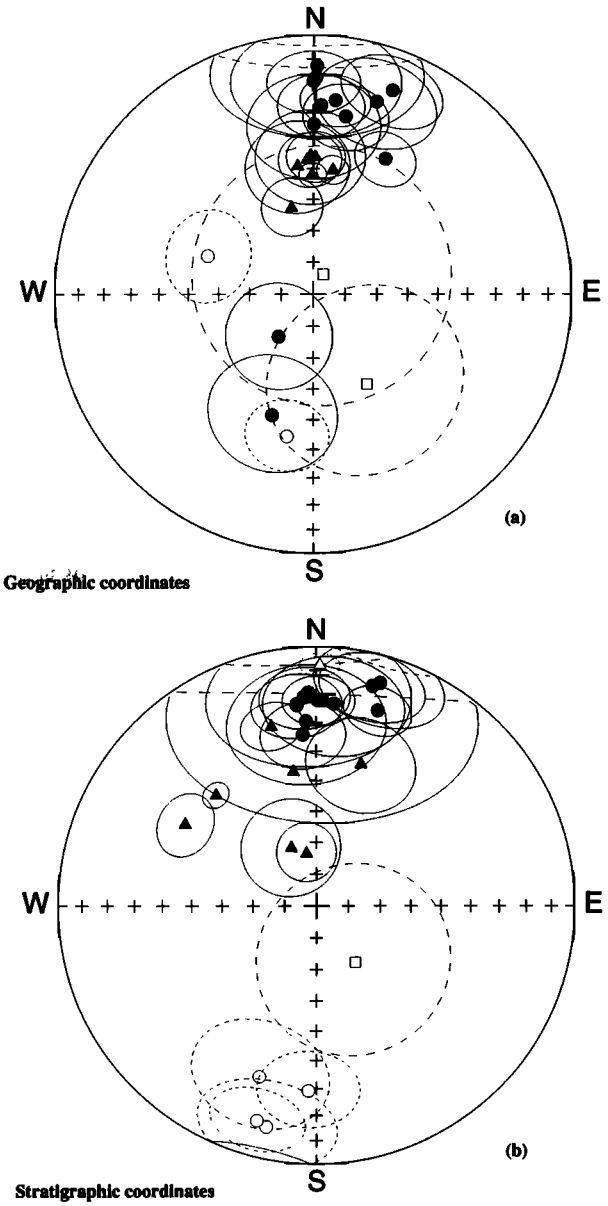


Fig. 6. Equal-area projection of site-mean directions of high-temperature component from all Cretaceous sites (a) before and (b) after bedding corrections. Squares correspond to "random" sites (compare Figure 5*l*), triangles to "remagnetized" sites (compare Figure 5*k*) and circles to remaining well-behaved sites (compare Figures 5*a*-5*j*), interpreted as primary, which show improved grouping direction after bedding correction. Solid (open) symbols are for directions in lower (upper) hemisphere.

coordinates (e.g., Fenghoushan group, site P23, [*Lin and Watts*, 1988, p.251]). The Jurassic Yanshiping formation from Qiangtang (North Tibet) block also appears to be remagnetized.

Achache et al. [1984] argued that they could distinguish between two separate tectonic subunits of the Lhasa (South Tibet) block. The boundary between the two was taken to follow the Nyainqentangla Range and the Jiali fault [*Tapponnier et al.*, 1986; *Armijo et al.*, 1986, 1989] (Figure 1). The southern subunit is more severely folded than the northern subunit and appears to have rotated clockwise by some 15° with respect to the northern subunit.

The southern part of the Lhasa (South Tibet) block has now been sampled by three separate teams [*Pozzi et al.*, 1982; *Achache et al.*, 1984; *Lin and Watts*, 1988], and it is of interest to compare these results, particularly those obtained in a small area to the NE of Lhasa near the farm village of Pengpo (30.0°N, 91.5°E). The sites sampled by Pozzi et al. and by Achache et al. can be subdivided in two subsets, one to the northwest and one to the northeast of Lhasa (Figure 9; for location see Figure 9*c* (inset) and the original papers, particularly, *Achache et al.* [1984] (Figure 2)). Each separate substudy (see Figure 9) provides a positive fold test and the six substudies together provide a larger-scale positive fold test. However, when one analyzes site results from the Pengpo area in detail (Figure 9*c*), some systematic differences appear. The sites of Achache et al. and Pozzi et al. come from precisely the same area (Figure 9*c*, inset); those of Lin and Watts come from

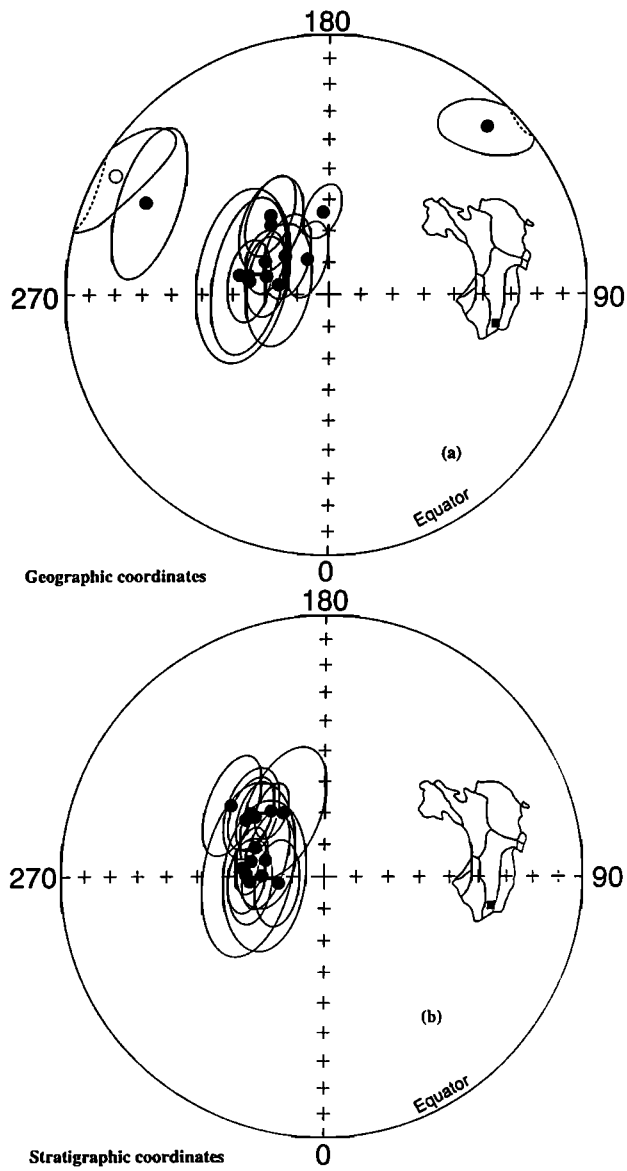


Fig. 7. Equal-area projection of characteristic Cretaceous site mean poles (*a*) before and (*b*) after bedding correction showing improved grouping direction after bedding correction. Square represents sampling locality (i.e., general area shown in Figure 2). Solid (open) dots are in the northern (southern) hemisphere.

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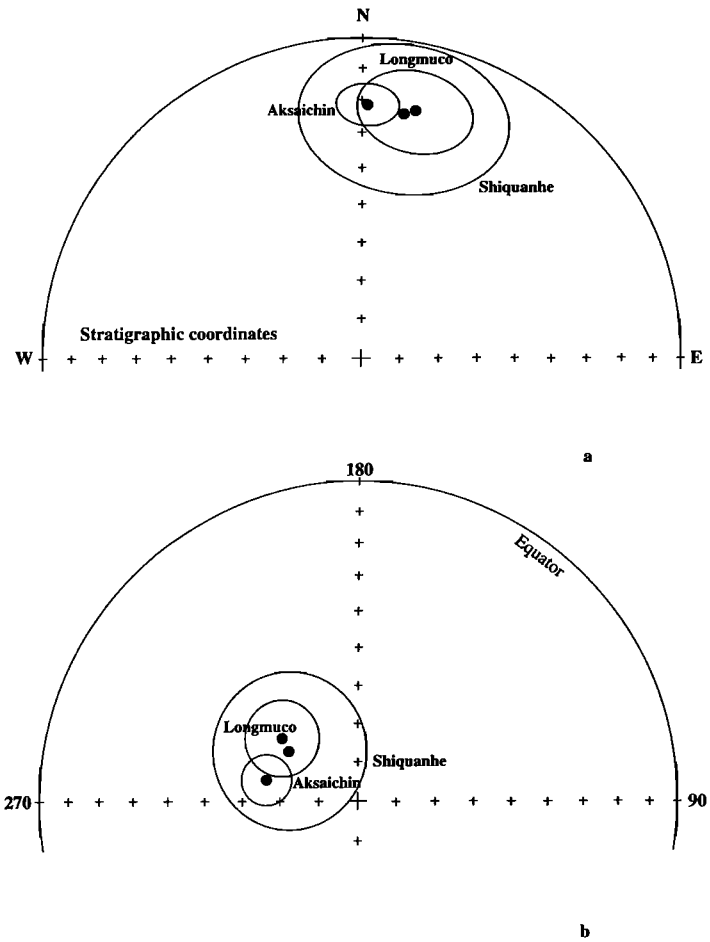


Fig. 8. Equal-area projection of (*a*) mean directions and (*b*) poles after bedding correction for the three areas discussed in this paper where Cretaceous results are available, showing good consistency. Note that the direction for Longmuco Cretaceous sites has been reversed for comparison.

the same general area, but their exact positioning with respect to the other sites is not known to us. The Achache et al. and Lin and Watts results appear to be in excellent agreement (Figures 9*b* and 9*c*). Those of Pozzi et al. tend to have more westward declinations and steeper inclinations after structural correction. The possibility of a PEF overprint has been suggested as a possible partial cause for the latter. The most removed direction comes from site 5, which was restudied in the field (V. Courtillot, unpublished field notes, 1981) and was found to be part of a wide, folded, channel structure. Bedding was hard to determine; alternate dip and strike measurements (V. Courtillot, unpublished field notes, 1981) yield a paleomagnetic direction with an inclination of 36° as opposed to 24°.

For poorly understood reasons, the sites sampled NW of Lhasa by Pozzi et al. (N=2) and Achache et al. (N=3) are in the same relative positions as the NE ones (Figure 9*b*), but the number of sites is too small to draw any significant inference. It is of course worrying that the sites sampled by Achache et al. in the northern part of the Lhasa (South Tibet) block have the same westerly declinations as those of Pozzi et al. in the south, although inclinations are lower (Figure 9*d*). Two courses are possible for use of the Lhasa (South Tibet) block Cretaceous data set. Either one can average the entire data set of 30 sites, yielding an average direction of D=347.7°, I=24.9° (α_{95}=5.2°) with a paleolatitude of 13.1°N [*Pozzi et al.*, 1982; *Achache et al.*, 1984; *Lin and Watts*, 1988] (Table 4), or one can decide that problems linked with overprints and structural corrections warrant rejection of the Pozzi et al. data. In that case, the average direction for 16 sites located on the southern portion of the Lhasa (South Tibet) block is D=355.5°, I=18.6° (λ=9.6°N, α_{95}=5.2° [*Achache et al.*, 1984; *Lin and Watts*, 1988] (Table 4)), significantly different from that from the northern portion of the Lhasa (South Tibet) block [*Achache et al.*, 1984]. This difference is negligible in terms of latitude (4°±7°) but amounts to a rotation of 18°± 8°, similar to that proposed by *Achache et al.* [1984].

Otofuji et al. [1989] sampled formations with diverse lithological facies and ages ranging from Silurian to Pliocene along the road from Chengdu in Sichuan, through Lhasa, to Kashgar in Xinjiang. Sites were separated by tens to hundreds of kilometers, and few sites of the same age were available (only hand samples were taken, from 4 to 10 per site). These data do not allow stability tests; precision parameters generally decrease upon bedding correction. Out of 11 sites, three were completely remagnetized, four were strongly deformed along the Yarlung-Zangbo suture, with declination almost aligned with the suture. The remaining sites, coming from the western part of the plateau (Shiquanhe to Quanshuihu) showed declinations ranging from 30° to -8°, similar to what we have found.

Otofuji et al. [1990] and *Huang et al.* [1992] have published Cretaceous results from the Markam area, in the far eastern part of the Tibetan plateau (Figures 1 and 12), on the Qiangtang (North Tibet) block. The results are consistent and those of Huang et al. pass both a fold and a reversal test. These authors deduce relatively high paleolatitudes, close to the present day ones (~30°N) and a large clockwise rotation of ~50°.

DISCUSSION

The new data and the review of previous work show that after over a decade of field work in Tibet, the space and time distribution of paleomagnetic data are still scarce. Whether in the western or eastern parts of either the Qiangtang (North Tibet) and Lhasa (South Tibet) blocks, Jurassic paleomagnetic directions tend to be remagnetized and Tertiary directions to be inconsistent [*Chen*, 1992; *Lin and Watts*, 1988]. A notable exception is the Lingzizong Paleocene volcanics in the Lhasa area which show consistent results (with fold and reversal test) and indicate a paleolatitude similar to the underlying Cretaceous strata [*Achache et al.*, 1984]. The best and most numerous results are obtained from Cretaceous rocks, mostly from red sandstones and in some cases limestones: they include studies from a 300 km section across the Qiangtang (North Tibet) and Lhasa (South Tibet) blocks in western Tibet (this study), a 200-km section across the Lhasa (South Tibet) block in the Lhasa area [*Achache et al.*, 1984; *Lin and Watts*, 1988] and a few localities in the Markam area in eastern Tibet [*Otofuji et al.*, 1990; *Huang et al.*, 1992]. We have plotted the Cretaceous latitudinal differences between the Eurasian reference [*Besse and Courtillot*, 1991] and available studied localities as a function of present longitude, i.e., along the strike of the Tibetan blocks and the major faults and sutures (Figure 10*a*) which separate them, together with the corresponding differences in declination (Figure 10*b*). The same information is displayed in a complementary fashion as a geographic plot of geomagnetic paleopoles (Figure 11). With the exception of the data from the Markam area within the great orogenic bend north of the Assam syntaxis, the first-order feature to emerge from these figures is the overall consistency of latitudinal changes or, equivalently, the fact that poles tend to follow a circle centered on Tibet. In

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TABLE 3. Summary of Cretaceous and Tertiary Paleomagnetic Poles of Tibet, India, and Pamir

Region	Site		Age	N/n	Pole		A95 (dp/dm)	Reference	Comment
	Lat	Lon			Lat	Lon			
South Tibet (Lhasa): Southern Part									
Shiquanhe Site 6*	32.5	80.1	T	6/9	64.9	222.1	6.8/12.7	*Otofuji et al.*[1989]	14 km, NFT
Jiangba Site 4*	32.6	80.4	K	6/11	15.3	300.9	3.2/6.2	*Otofuji et al.*[1989]	5 km, no FT
Mentu Site 25*	31.2	80.9	T (Pa)	5/10	38.5	9.9	17.7/25.9	*Otofuji et al.*[1989]	11 km, no FT
Laangco Site 29*	30.7	81.4	K	3/11	25.8	339.3	11.5/22.9	*Otofuji et al.*[1989]	0 km, no FT
Yamu Site 35*	29.4	87.7	K/T	5/11	8.4	13.6	18.7/33.0	*Otofuji et al.*[1989]	23 km, NFT
Lhasa	30.0	91.0	K	8	71.2	288.4	7.9	*Achache et al.* [1984]	PFT
Lhasa	30.0	91.0	T (Pa)	8	71.4	299.8	6.4/11.9	*Achache et al.* [1984]	no FT
Lhasa	29.9	91.0	K	6	64	348	5.6/9.5	*Westphal et al.* [1983]	PFT
Lhasa	29.7	91.2	K	7	68	340	10	*Pozzi et al.* [1982]	all but one site as above
Lhasa	29.9	91.2	K	8	68	279	3.5/6.9	*Lin and Watts* [1988]	PFT
Basu	30.1	96.9	K	4	75.8	264.5	17.0/30.9	*Otofuji et al.* [1990]	No FT
South Tibet (Lhasa): Norhern Part									
Shiquanhe	32.7	80.2	K	3	67.7	234.2	13.1/24.5	this study	no FT
Qelico	31.7	90.9	K	7	74	318	11.1/19.1	*Lin and Watts* [1988]	no FT
Naqu	31.5	91.0	K	9	78	282	4.0/6.9	*Lin and Watts* [1988]	no FT
Anduo	31.0	92.0	K	6	63.5	325.4	6.5	*Achache et al.* [1984]	PFT
North Tibet (Qiangtang)									
Aksaichin	35.0	79.7	K	7	66.3	256.5	3.7/7.0	this study	PFT
Ritu Site 37*	33.3	79.7	K	5/10	55.7	199.4	1.9/3.5	*Otofuji et al.*[1989]	40 km, No FT
Bangonco Site 21*	33.5	79.9	J/K	4/5	65.7	214.1	13.9/25.3	*Otofuji et al.*[1989]	85 km, NFT
Quanshuihe Site 19*	34.8	80.1	K	5/7	49.2	273.1	6.1/12	*Otofuji et al.*[1989]	225 km, NFT
Longmuco	34.5	80.4	K	4	64.4	231.3	7.2/13.5	this study	no FT
Erdaogou	34.6	92.4	T(Pa-E)	15	61	253	3.8/7.4	*Lin and Watts* [1988]	no FT
Erdaogou	34.6	92.8	T(Pa-E)	12	55	201	3.2/5.6	*Lin and Watts* [1988]	no FT, discordant with above results
Markam	29.7	98.4	Kl	12	40.6	170.5	13.0	*Huang et al.* [1992]	PFT
Markam	29.7	98.6	K	5	48.5	175.9	9.5	*Otofuji et al.*[1989]	no FT
Markam	29.7	98.7	Km	10	58.7	173.2	11.1	*Huang et al.* [1992]	PFT
India									
N. Kirthar	30.1	66.6	T	4 (n=87)	53.9	315.5	8.6	*Klootwijk et al.* [1981]	no FT
C. Lorals and S.Sulaiman	30.1	69.9	T	4 (n=106)	59.9	200.8	4.7	*Klootwijk et al.* [1981]	no FT
Chitral and W Salt Range	35.3	72.2	T	6 (n=424)	42.5	305.2	14.6	*Klootwijk et al.* [1984*a*, 1986*b*]	no FT
Kashmir and Jammu	34.1	74.4	T	7	64.3	199.0	5.5	*Klootwijk et al.* [1983, 1986*a*] *Bossart et al.* [1990]	no FT
Ladakh	34.4	76.7	T	8 (n=208)	55.3	252.8	8.5	*Klootwijk et al.* [1979, 1984*b*]	no FT
Krol belt/Less Himalaya	30.2	78.4	T	2	66.3	197.5	-	*Klootwijk et al.* [1982]	no FT
Kali, Nepal	28.7	83.7	T	6 (n=237)	57.2	276.9	10.3	*Klootwijk and Bingham* [1980]	no FT
Dingri	29.0	87.0	T (Pa)	4	49.8	309.7	4.2/8.4	*Besse et al.* [1984]	PFT
Pamir									
Locality 10	38.0	69.5	T (Pa)	n=39	36.9	330.2	4.2/7.6	*Bazhenov and Burtman* [1986]	na
Locality 8	38.7	70.5	T (Pa)	n=31	63.9	26.9	10.8/12.6	*Bazhenov and Burtman* [1986]	na
Locality 1	39.5	74.0	T (Pa)	n=28	50.5	174.8	6.1/10.1	*Bazhenov and Burtman* [1986]	na
Locality 2	39.5	73.9	T (Pa)	n=27	54.7	185.1	5.2/8.9	*Bazhenov and Burtman* [1986]	na

N and n correspond to the number of hand samples and specimens respectively, dp/dm are the incertainties of mean poles when they are computed from mean directions, and the figure in the comment column refers to the distance from the Indus-Zangbo suture (in km). PFT, positive fold test; NFT, negative fold test; no FT, without fold test. T, Tertiary; K, Cretaceous, Pa-E, Paleocene-Eocene, K1, Lower Cretaceous, Km, Middle Cretaceous.

*: For sites from *Otofuji et al.* [1989].

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TABLE 4. Selected Average Cretaceous Paleomagnetic Directions and Poles of Tibet

Block	N	Ds (ΔD)	Is (ΔI)	λ (Δλ)	Pole Lat	Pole Lon	A95 (dp/dm)	Reference
80°E Longitude Traverse								
Qiangtang (Aksaichin)	7	1.3 (7.1)	21.8 (6.6)	11.3 (+3.8 -3.6)	66.3	256.5	3.7/7.0	this study
Qiangtang (Longmuco)	4	12.4 (13.8)	22.2 (12.8)	11.5 (+7.8 -6.7)	64.4	231.3	7.2/13.5	this study
Lhasa (Shiquanhe)	3	9.8 (25.1)	23.5 (23.0)	12.3 (+15.5 -12.0)	67.7	234.2	13.1/24.5	this study
Mean	14	6.3 (5.2)	20.6 (5.1)	10.6 (±5.1)	66.2	245.0	5.1	this study
90°E Longitude Traverse								
South Lhasa (Lhasa)	16	355.5 (5.5)	18.6 (5.2)	9.6 (+2.8 -2.8)	69.1	283.5	2.8/5.4	*Achache et al.* [1984] and *Lin and Watts* [1988]
North Lhasa (Anduo)	7	337.8 (6.1)	27.0 (5.9)	14.3 (±5.9)	63.4	326.2	5.9	*Achache et al.* [1984]
Lhasa (North+South)	30	347.7 (5.7)	24.9 (5.2)	13.1 (+3.1 -3.0)	69.6	307.5	3.0/5.6	*Pozzi et al.* [1982], *Achache et al.* [1984] and *Lin and Watts* [1988]
Markam								
Qiangtang (?)	27	48.4 (7.0)	50.7 (6.0)	31.4 (±6.0)	48.6	173.5	6.0	*Otofuji et al* [1989] and *Huang et al.* [1992]
Eurasia								
	29				76.3	202.6	2.2	*Besse and Courtillot* [1991]

See Figures 1 and 12 for site locations and Figures 10 and 12 for display. D_S, I_S, λ: declination and inclination after bedding correction and corresponding paleolatitudes, given with 95% uncertainties (in parentheses). In the cases of the Qiangtang (Aksaichin), Qiangtang (Longmuco), Lhasa (Shiquanhe) and Lhasa (N+S) data, average directions are calculated first and mean poles with uncertainty (dp/dm) are calculated next. In the other cases, means of poles are calculated with uncertainty A_{95} and directions are recalculated from them. Corresponding formulas for uncertainties after *Butler* [1991].

contrast, these poles are distributed along this circle, corresponding to a scatter in declination which is visible at all scales (e.g., Figures 7, 9*c*, 9*d* and10-12).

At the time of deposition of these rocks, circa 100 Ma, India was moving northward, still a rather long way from initial collision [e.g., *Patriat and Achache*, 1984; *Jaeger et al.*, 1989], whereas Eurasia was essentially stationary [*Besse and Courtillot*, 1988, 1991]. The Cretaceous paleolatitudes from Tibetan sites are therefore consistent with the view that Tibet formed the stable southern margin of Eurasia with a general east-west trend at about 10°N latitude. The Domar and Lhasa areas, now near 34° and 30°N, respectively, both stood near 10°N latitude, implying over 2000 km of northward displacement (Figure 10*a* and Table 4), most likely since the time of the collision, i.e., in the last 60 Ma.

On the other hand, Figure 11a shows that the various poles do not coincide, which means that the blocks have suffered internal deformation. That such deformation must have occurred is obvious from small- and large-scale Cenozoic tectonic strain within the plateau (Figure 1) [e.g., *Tapponnier et al.*, 1981, 1986; *Armijo et al.*, 1986, 1989; *Chang et al.*, 1986, 1989; *Avouac*, 1991; *Bourjot*, 1991; P. Matte et al., unpublished manuscript, 1993] even though the exact amounts of finite deformation are not yet known with sufficient confidence. Paleomagnetic evidence for regional internal deformation of the Lhasa (South Tibet) block at the 100 km scale was given by *Achache et al.* [1984]. Because folding had been stronger in the southern Lhasa (South Tibet) block, closer to the Yarlung-Zangbo suture, the northern data were believed to be more representative of the South Tibet (Lhasa) block as a whole.

The data from western Tibet given in this paper can be compared to the Lhasa data. Our data come from three general areas which lie very close (50 to 5 km) from major tectonic zones. The Aksaichin and Longmuco sites straddle the left-lateral Gozhu-Longmuco strike-slip fault system, although in Aksaichin the strata form gentle concentric folds. The Shiquanhe sites are north of but close to major thrusts and not far from the great right-lateral Karakorum fault (Figure 2). One might have suspected that declinations could therefore have been affected by local shear, with clockwise rotations in the former and anticlockwise rotations in the latter case. The fact that the poles and directions overlap (Figure 8), even though the uncertainty in the Shiquanhe pole is rather large (α_{95}=23°), implies that to first order, local differential strain was not large and that this small (100 by 300 km) block of western Tibet behaved rather rigidly. Actually, our results only imply that strain was not large enough to be determined by paleomagnetism. It is clear that some shortening occurred since the Cretaceous strata are, sometimes strongly, folded (Matte et al., unpublished manuscript, 1993). Also, continued extrusion of western Tibet along the antithetic Karakorum and Longmuco faults (and the Altyn Tagh fault farther north) cannot have been a completely rigid process. It is interesting that the azimuthal difference between the paleomagnetic directions in western Tibet (Lhasa and Qiangtang (South and

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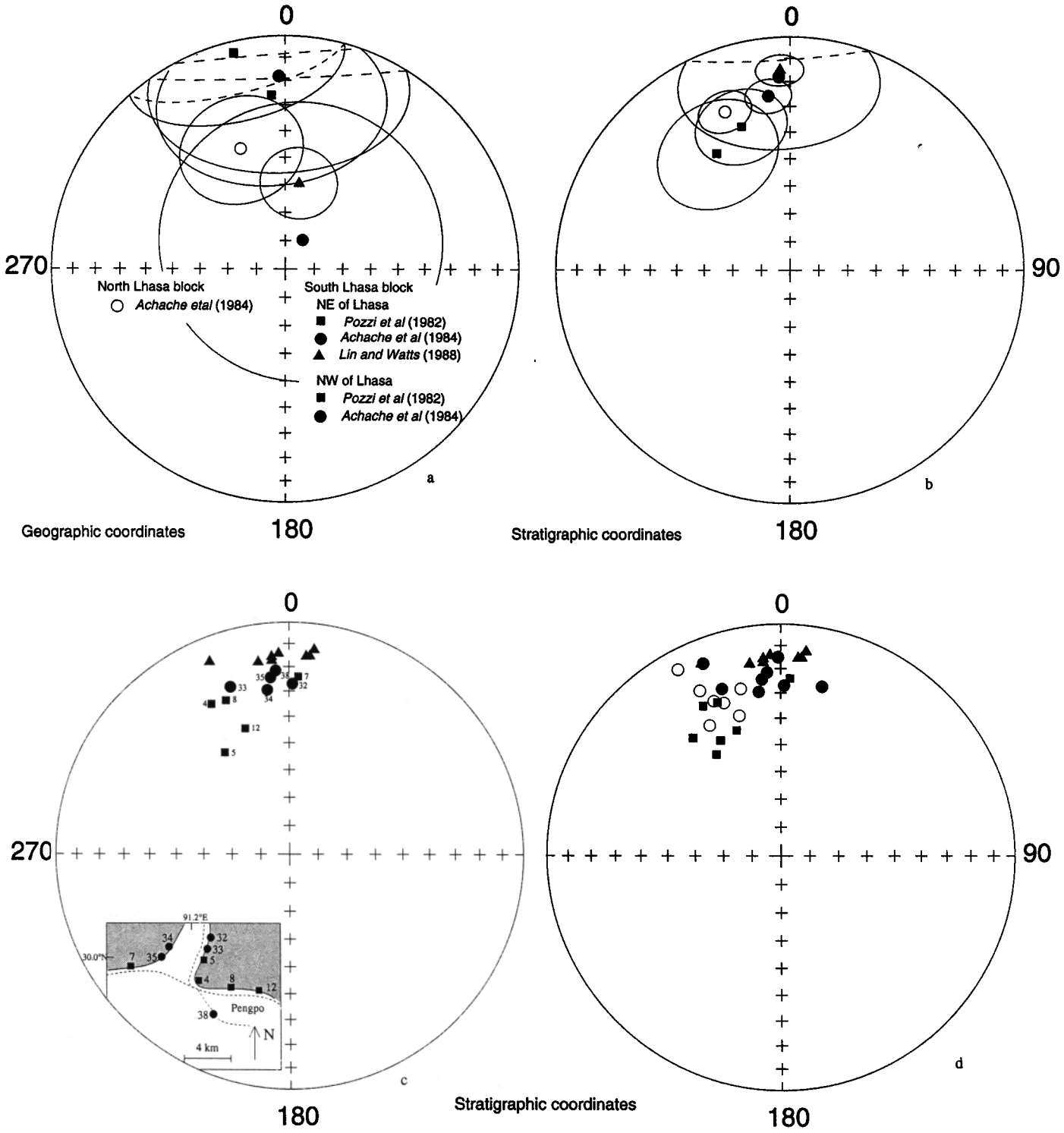


Fig. 9. Equal-area projection of paleomagnetic results for the Lhasa block along the 90°E traverse [*Pozzi et al.*, 1982; *Achache et al.*, 1984; *Lin and Watts, 1988*]. (*a*, *b*) Means have been recalculated from these three studies separately for the northern part of the Lhasa block, and in the southern part of the Lhasa block for sites to the NW and NE of Lhasa. (*c*) Details of site mean results for the Pengpo area, NE of Lhasa, with site locations in the inset. (*d*) All individual site-mean results. Symbols are the same in all parts of the figure, and are shown in Figure 9*a*. Note that all inclinations are downward including those with open symbols. See text for explanations.

North Tibet, respectively) blocks) (this study) and in the northern Lhasa (South Tibet) block in eastern Tibet (29°±8°) is similar to the azimuthal difference in tectonic fabric, as visible for instance in the trend of the Bangong-Nujiang suture (about 25°). This would be compatible with the idea that the suture was more or less linear prior to the collision and acquired its curvature subsequently, a form of oroclinal bending already proposed, with larger amplitude, for the

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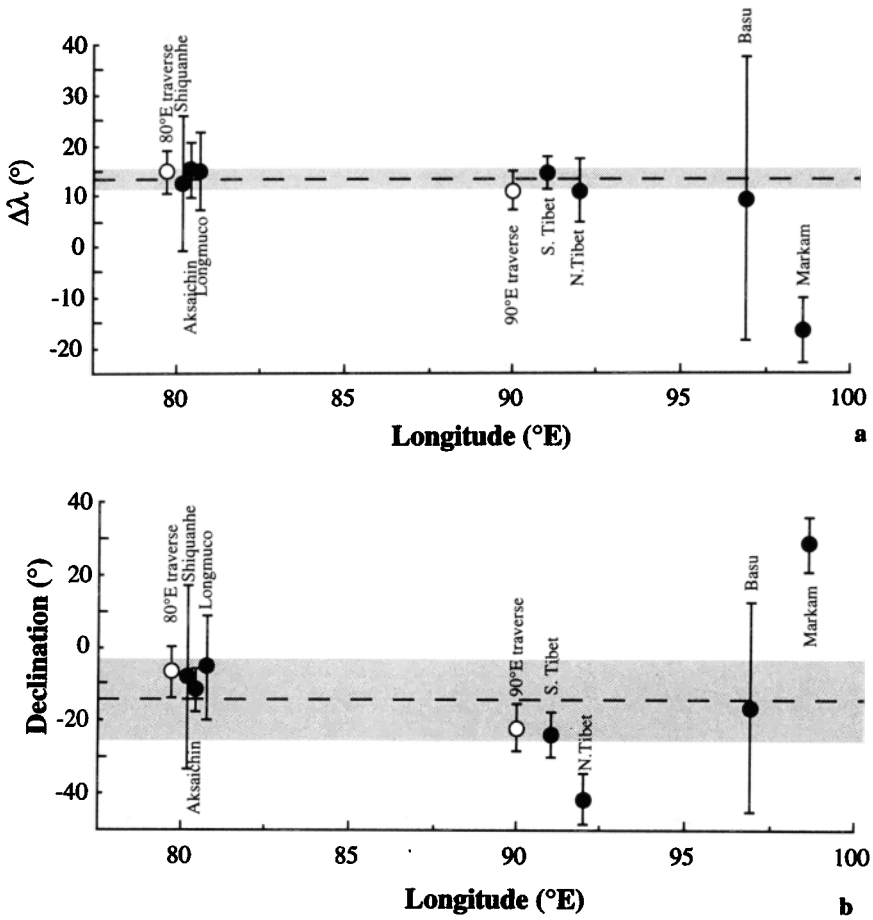


Fig. 10. Cretaceous (*a*) latitudinal and (*b*) declination differences between the Eurasian reference [*Besse and Courtillot*, 1991] and available studied localities in Tibet as a function of longitude (solid symbols). Open symbols correspond to averages of the 80°E traverse and 90°E traverse results (see Table 4 for values and references). Shaded bands correspond to overall mean (excluding the Markam data) with standard deviation (*N*=6 values).

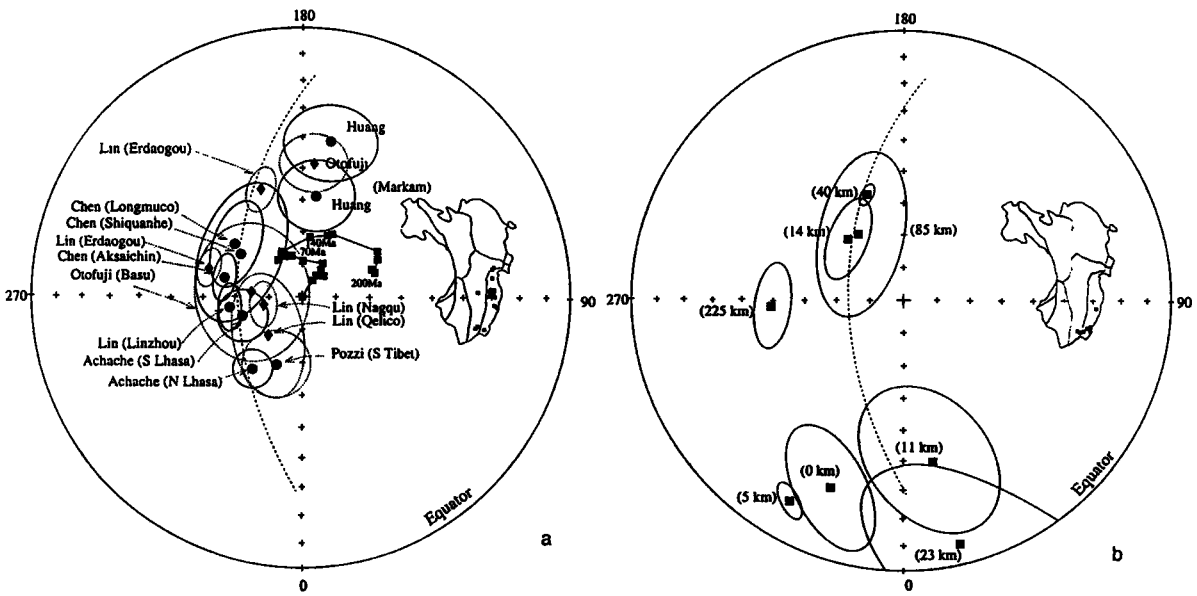


Fig. 11. Equal-area projection of Cretaceous paleomagnetic poles of Tibet with site locations. See Table 3 for values and references. (*a*) Data from main plateau (circles are for poles with a positive fold test, diamonds for poles without a test; dashed confidence intervals correspond to less reliable or superceded results); (*b*) results from *Otofuji et al.* [1989] (J/K to lower Tertiary) for western Tibet showing large scatter (the distance from the Indus-Zangbo suture is given in parentheses).

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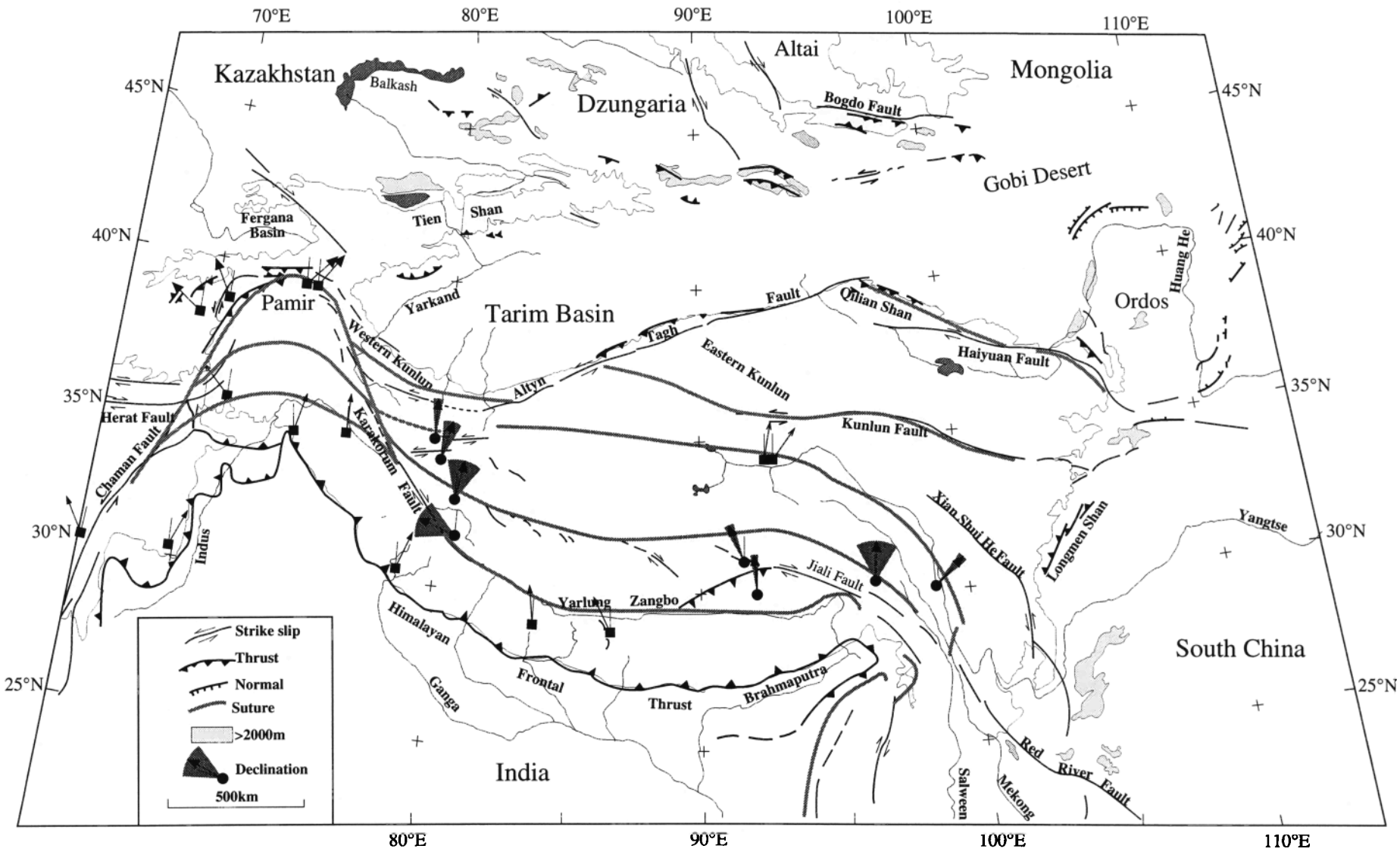


Fig. 12. Tectonic map of Tibet and surrounding areas with sites (circles) where Cretaceous paleodeclinations are available (arrows and 95% uncertainties; see Table 4), showing the extent of differential rotations. Tertiary magnetizations or remagnetizations are also shown for comparison (squares). The thin lines next to arrows are north reference for declination (not exactly parallel to local north because of map projection).

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southern boundary of the Lhasa (South Tibet) block with the Indian indenter [*Klootwijk et al.*, 1985].

However, the 29° declination difference between Domar (80°E longitude) and North Lhasa (90°E longitude) is not very much larger than the apparent declination difference between the northern and southern portions of the Lhasa (South Tibet) block (18°), or the scatter within individual substudies such as those shown in Figures 6 (for the western traverse) and 9 (for the eastern traverse). The corresponding horizontal scales are 1000 km, 100 km and 1 to 10 km, respectively. Hence the data density is not sufficient to confidently separate local rotations about vertical axes related to segmentation by small-scale (1 to 100 km) sets of faults from large scale (1000 km) "oroclinal" bending. Some of the data collected close to the Indus-Zangbo suture by *Otofuji et al.* [1989] show extreme examples of rotated declinations probably due to strike-slip faulting (Figure 11*b*) and are generally consistent with the paleolatitudes and large declination differences shown by the Cretaceous data (Figure 11*a*).

The data of *Otofuji et al.* [1989] and *Huang et al.* [1992] in the Markam area of easternmost Tibet deserve special attention. The large clockwise rotation is not surprising, because the sampling sites belong to a zone of considerable deformation, shear and extrusion around the northeastern corner of the Indian indenter [*Tapponnier et al.*, 1982, 1990; *Peltzer et al.*, 1989]. The paleolatitudes, on the other hand, do not appear to be consistent with the other Cretaceous data from Tibet. They are close to the present day latitudes, approximately 21°±10° farther north than expected for the Lhasa (South Tibet) block in the mid-Cretaceous and 22°±11° farther north than expected for the Qiang-tang (North Tibet) block (Figure 10*a*). *Huang et al.* [1992] interpret this observation in terms of a giant right-lateral transpressional zone advocated by *Dewey et al.* [1988]. This interpretation assumes east-west compression, which does not seem compatible with the hypothesis of plane strain (which underlies the bookshelf faulting model on which the Dewey et al. idea is based; see also *England and Molnar* [1990]). We can offer alternate interpretations. If the paleolatitudes do reflect the geographical location of the Markam area in mid-Cretaceous time, the crustal slice on which Markam now stands might have belonged either to South China or to a block located within northern Tibet, possibly farther west than its present location. It might even have been part of the northwestern wedge of Indochina, which was apparently extruded to the southeast by the penetration of India into Asia [*Tapponnier et al.*, 1986, 1990; *Peltzer and Tapponnier*, 1988; *Yang and Besse*, 1993]. Indeed, large relative north-south motions can be expected near the eastern edge of the Assam syntaxis. Alternatively, the (lower?) Cretaceous (or upper Jurassic ?) rocks of Markam could have been almost completely remagnetized in fairly recent times, subsequent deformation by shear leading to the observed clockwise rotation and positive fold test.

Because of the complexity, intensity and duration of deformation in Tibet, it is difficult to propose an unambiguous paleogeographic reconstruction of the Tibetan blocks at this stage. Relying on the inclination data however, it is probably safe to argue that the southern margin of Tibet (i.e., of Eurasia) occupied a position near 7°±6°N latitude (assuming Lhasa was close to 10°N) with a roughly eastward trend in mid-Cretaceous time. Taken at face value, the bending of the Bangong-Nujiang suture and the change of paleodeclinations from east to west would imply that, starting from an initial trend of about 100°E, the arcuate shape and rotations resulted from the collision process.

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